1 System identification

Given the importance of oscillation frequency for muscle selection and periodicity for muscle power, we evaluate a system identification strategy that seeks to align the analytical frequency response derived from system matrices to the empirical frequency response measured from the brain.

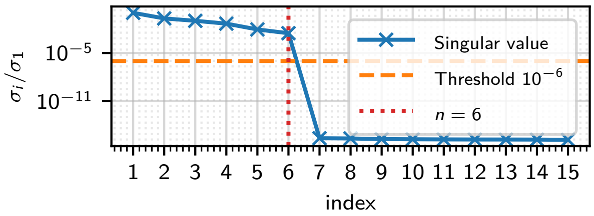

Figure 1: Normalised singular values σ_i/σ_1 of the block Hankel matrix constructed from 55 noise-free Markov parameters (20 block rows, 30 block columns, $p = 2$ inputs, $q = 16$ outputs). The orange dashed line marks the relative threshold 10^{-6}; singular values below this level are indistinguishable from numerical zero under exact noise cancellation. The red dotted line indicates the selected model order $n = 6$. Source

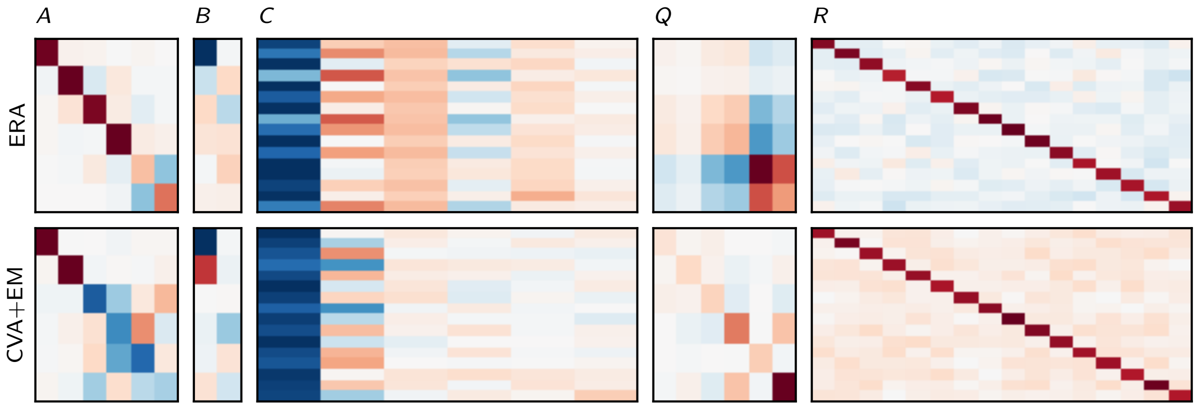

Figure 2: System matrices (A, B, C, Q, R) identified from the Brain simulator with model order $n = 6$, $p = 2$ inputs, and $q = 16$ outputs, using $N = 5000$ samples for both methods. Top row: Eigensystem Realisation Algorithm (ERA). Bottom row: Canonical Variate Analysis initialisation followed by Expectation–Maximisation (CVA+EM). Each column shares a common colour scale across rows to enable direct comparison between methods. Source

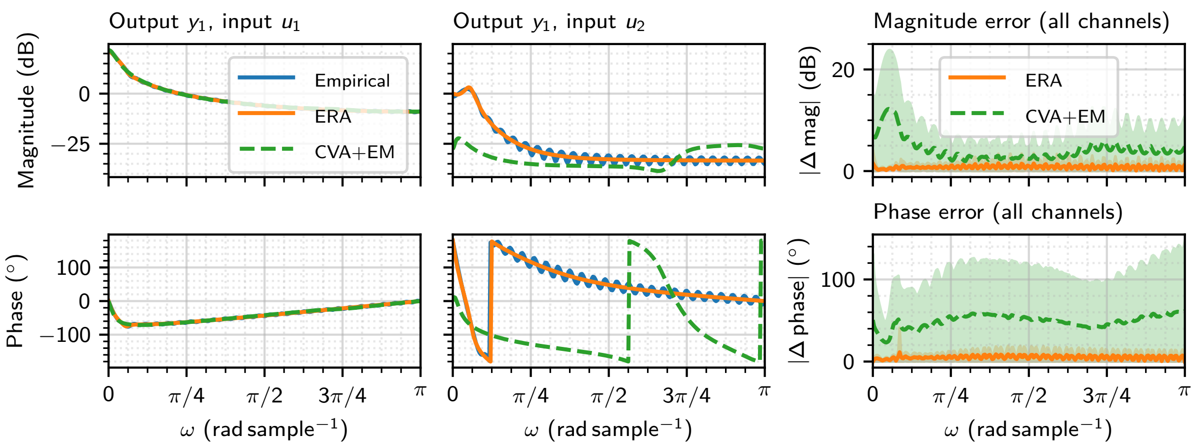

Figure 3: Frequency-response comparison for ERA and CVA+EM identified models. Left and centre columns: Bode magnitude (top) and phase (bottom) for output y_1 driven by inputs u_1 and u_2 respectively, comparing the empirical response (discrete Fourier transform of 55 noise-free Markov parameters, zero-padded to 512 points) against the analytic response $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ from the ERA model (order $n = 6$) and the CVA+EM model (order $n = 6$); the two columns share a common y-axis scale. Right column: absolute magnitude error (top) and absolute phase error (bottom), shown as mean $\pm$ 1 s.d. across all 32 output–input channels for each method. Source